

Interação em Linguagem Natural (MI/MEI/MSI)
Tecnologia da Linguagem (DCC & MCC)
2025/2026

Project Type 2

(this Project program has 4 pages)

Reading comprehension

With a pedagogical purpose, in this project you will develop a program with two aims.



On the one hand, the program results from training a neuronal model of the Transformer class so that it becomes suitable for a reading comprehension task. The task's input includes a multiple-sentence context, a question to be answered from that context, and a candidate answer to that question. The output is a binary 0/1 where 0 stands for the candidate answer being wrong, and 1 stands for it being correct. Examples:

Context	Question	Candidate Answer	Label
A stranger in town meets pretty young Susan Martinez De La Cruz and accompanies her to a barbecue , where wealthy Jason Carberry is saying a few words for the recently departed Robin Randall , a citizen who got shot . Jason objects to the stranger's presence , being Susan's guardian and protective of her . He challenges him to a shootout , but the stranger pulls his pistol before Jason's can even clear the holster . Calaveras Kate , a saloon singer who's in love with Jason , is relieved when the stranger declines to pull the trigger. ...	Who is brawling and what causes it to end?	Jason and Susan	0

The Finnish government and major creditors of bankrupt shipyard Waertsilae Marine Industries Oy agreed in principle to form a new company to complete most of the troubled shipyard's backlog of 15 ships. The new company will attempt to limit the shipyard's losses, participants said. "The situation is that the bankruptcy court will get out of the shipbuilding business. Everything will be taken over by the new company," said Christian Andersson, executive vice president of Oy Waertsilae, former parent of Waertsilae Marine...

After ownership is finalized, what will be the offer to the Subcontractors?

1

A settlement
and swift
transition

In addition, having trained the model as indicated above, the program allows you to input a pair consisting of any context, any question and any candidate answer, for which it will return a 0/1 answer, as determined by the model.

1. Specification

Hardware

Google Colab.

Software

Python, Pytorch, HF.

Models

The two neural networks to be developed must be instances of the RoBERTa family of the Transformer architecture.

Approach

Tuning a pre-trained model.

Pre-trained model for the English language to use:

<https://huggingface.co/roberta-base>

Dataset

MultiRC dataset for Multiple Sentences Reading Comprehension:

https://huggingface.co/datasets/aps/super_glue/viewer/multirc

Given hyperparameters

Batch of size 8.

Training for 1 epochs.

Programme output

The programme ensures that a local copy of the tuned model is saved.

Prints the values of the evaluation of task performance (accuracy).

Code

The programme to be submitted for assessment in the course is a Jupyter notebook capable of running on Google Colab with HF.

This programme must implement the tuning of the adopted language model (RoBERTa) in relation to the intended task (MultiRC).

This programme must also implement a function `test(passage, question, answer)`, which accepts as input a context, a question that must be answered given that context, and a candidate answer to that question, and returns as a result the answer which is the output of the model that has been tuned for MultiRC, for that input.

2. Software development

Groups

This project is solved by groups with exactly 2 members.

Groups registration

Through moodle page of this course.

Support for solving the project

The means of pedagogical support for the students of this course continue to be available, which have been available since the beginning of the course, and which can and should be used to support the resolution of this project. Recall they are the following:

- contact with teachers at the end of classes throughout the semester
- individual and personalized office hours for students throughout the week
- course forum on moodle, with access by all students
- courses news on moodle

Given the nature of the task to be carried out and the complex context of the code in which possible difficulties arise, clarifications on the resolution of the project must be obtained through these support means, **not being answered through email messages to the individual addresses of the teachers.**

3. The evaluation component

Elements to be submitted by the students for evaluation

To submit your group's resolution for evaluation, deliver a single .zip FILE, (for example, projectType2Grupo564.zip) that results from compressing the folder with:

- the code files developed,
- the file with the report.

Each of the code files, in turn, must contain in the first lines, as comments, information about the group number and the number and full name of each group member who worked on the project, as exemplified below:

```
# 2025-2026 Interação em Linguagem Natural  
# Grupo 12  
# 62345 Felismina Felisbela  
# 78766 Aniceto Ambrósio
```

Implementation report

The implementation report must not exceed three pages, is named relGrupoN.pdf (where N is the group number) and must be in .pdf format. Reports in other formats will be ignored. It must contain the following sections:

1. Group number
2. Number and full name of each group member
3. Presentation of the essential lines of the structure and operation of the software developed
4. Hyperparameter values
5. Performance values in training, validation and testing
6. Indication of the features that remained to be implemented (if applicable)
7. Indication of known errors (if applicable)
8. Indication of what each member of the group accomplished for the resolution of the project, and the proportion in which each one contributed to the number of hours applied in its resolution.

Classification

The individual classification of the project resolution is weighted by the individual performance of each element of the group in the oral discussion of the resolution (if applicable).

Academic integrity

As a future professional, you are expected to have an irreproachable attitude in deontological terms. Take the greatest care to respect and enforce the law on computer crime.

Students detected in a situation of fraud or partial or total plagiarism — plagiarists and plagiaries, with or without the intervention of intermediaries — in any evaluation component will have this evaluation element canceled and will be subject to

disciplinary proceedings. Don't want to have to show your diploma to a future employer with such an incidence recorded.

There can and should be mutual help between students, through the discussion of applicable methods and algorithms. It is however the sole responsibility of each group to take steps to protect their code from being plagiarized.

In the evaluation process, support software will be used in the detection of plagiarism that compares the answer of each group with each of the answers of the other groups.

Form and date of delivery

Students deliver a .zip file, which results from compressing the folder with the code files and report.

The only DELIVERY METHOD is through the discipline website
Any delivery in another form will not be considered for evaluation.

To be evaluated, the result must be submitted by the DEADLINE of Monday, June 15, 2026, 23:00 (Lisbon time).

Any submission or resubmission after this deadline will not be considered for evaluation.